

Chat GBT Prompts during the project

Coding Side

I used AI Copilot extensively in order to debug my code if It wasn't running properly, I started to use grok aswell much more since it has a python running code built into its interface which allows me to run and debug code before putting it into my coding platform software.

For each Kernel in my code other than the data extraction and descriptive statistics where I decide to use past work to help me code what I wanted I must have used the debugging tool at least 2 times.

For the Hasio regressions and the use of Newy west standard errors I used the Grok AI tool heavily in order to help me build the regressions and interpret the results.

I good tool that I find AI to be most useful for in my coding is during the result outputting section, I find this work tedious and also with the default output of descriptive statistics when I'm not using the package "data wrangler" is: hard to interpret and un-neat. In order to save time I would almost always just copy my code into ChatGPT and ask it to write the standard:

```
print(f"\n{name}")
print("-" * 50)
print(f"F-statistic : {res['F statistic']:.3f}")
print(f"p-value      : {res['p-value']:.4f}")
print(f"Restrictions: {res['q restrictions']}")
```

or

```
print("Kept:", out["kept"])
print("Dropped (var, VIF at drop time):", out["dropped"])
print(out["vif_final"])
print(out["result"].summary)
```

This would save me a lot of time as its normally where I would make mistakes anyway in my code since I wouldn't really pay any attention when I was typing it out.

All my latex tables we made using ChatGBT inorder to save time, I would simply just cop directly form my VScode the table and ask it to code It for me. Example:



code this table into latex for me

Here is your table coded cleanly in **LaTeX**, matching what is shown (lags 0–9, 7 variables). I use

booktabs for professional formatting.

```
latex
\begin{table}[htbp]
\centering
\caption{Lagged coefficients}
\label{tab:lagged_coefficients}
\begin{tabular}{lrrrrrrrr}
\toprule
Lag
& Bank loans / GDP
& Liquid liabilities
& $\ln(\text{CPI})$
& $\ln(\text{GDPpc})_{t-5}$
& $\ln(\text{Labour productivity growth})$
& $\ln(\text{Trade openness})$
& Market cap / GDP \\
\midrule
0 & -0.064 & -0.176 & 0.128 & 0.033 & 0.796 & 0.107 & 0.011 \\
1 & -0.086 & -0.154 & 0.135 & 0.028 & 0.726 & 0.102 & 0.023 \\
2 & -0.114 & -0.123 & 0.137 & 0.027 & 0.549 & 0.086 & 0.013 \\
3 & -0.139 & -0.079 & 0.137 & 0.028 & 0.375 & 0.055 & 0.001 \\
4 & -0.154 & -0.011 & 0.137 & 0.031 & 0.192 & 0.016 & -0.012 \\
5 & -0.155 & 0.052 & 0.135 & 0.032 & 0.006 & -0.008 & -0.017 \\
6 & -0.164 & 0.067 & 0.126 & 0.030 & -0.050 & -0.015 & -0.031 \\
7 & -0.168 & 0.066 & 0.117 & 0.026 & -0.046 & -0.021 & -0.022 \\
8 & -0.165 & 0.057 & 0.110 & 0.023 & -0.038 & -0.028 & -0.019 \\
9 & -0.160 & 0.040 & 0.107 & 0.018 & -0.022 & -0.040 & -0.029 \\
\bottomrule
\end{tabular}
\end{table}
```

remove cpi trade openness and labour productivity

Here is the **updated LaTeX table** with **CPI, trade openness, and labour productivity removed**, keeping only the remaining variables.

```
latex Copy code

\begin{table}[htbp]
\centering
\caption{Lagged coefficients (selected variables)}
\label{tab:lagged_coefficients_reduced}
\begin{tabular}{lrrrr}
\toprule
Lag
& Bank loans / GDP
& Liquid liabilities
& $\ln(\text{GDPpc})_{t-5}$
& Market cap / GDP \\
\midrule
0 & -0.064 & -0.176 & 0.033 & 0.011 \\
1 & -0.086 & -0.154 & 0.028 & 0.023 \\
2 & -0.114 & -0.123 & 0.027 & 0.013 \\
3 & -0.139 & -0.079 & 0.028 & 0.001 \\
4 & -0.154 & -0.011 & 0.031 & -0.012 \\
5 & -0.155 & 0.052 & 0.032 & -0.017 \\
6 & -0.164 & 0.067 & 0.030 & -0.031 \\
7 & -0.168 & 0.066 & 0.026 & -0.022 \\
8 & -0.165 & 0.057 & 0.023 & -0.019 \\
9 & -0.160 & 0.040 & 0.018 & -0.029 \\
\bottomrule
\end{tabular}
\end{table}
```

This was incredibly tie effective and also allowed me to also improve the presentation of my project as latex tables are much nicer to study than the standard Python output.

Writing Side

I didn't use ChatGPT often during my writing side but for huge tables I would ask it to pick out statistics and findings that stood out and would make more sense to talk about so I wouldn't have to spend too much time analysing my output as the AI would pre select the most important findings and all I would have to do would be to visually check them and make sure what he had given me wasn't an error.

I would also have it explain to me some concepts also, this was during also suring my revision for the panel data exam. Example:

what is a newy west standard erros and how does it help me against serial correaltion??

Great question — this is exactly the sort of thing examiners ask because they want to see whether you understand **what Newey–West actually does** (and what it does *not* do).

I'll explain it **conceptually, mathematically (lightly), and in terms of your results**.

1. What Newey–West standard errors are (plain English)

Newey–West (NW) standard errors are:

Standard errors that remain valid when regression residuals are autocorrelated and heteroskedastic.

They do **not** change your coefficient estimates.

They only change **how uncertain those estimates are**.

So:

- coefficients → same as OLS
 - standard errors → corrected
 - t-statistics / p-values → corrected
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